Assignment

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All valuations in this proof are nontrivial discrete valuations. Semi-abelian reduction means semistable reduction: the connected special Néron fiber is an extension of an abelian variety by a torus, without requiring the torus to split. All automorphisms of abelian varieties used for twisting preserve the origin.

The proof uses the supplied one-fiber theorem and the verified finite-family degree formula. It adds constructions that attain the full wild exponent independently at each of the prescribed fibers, and a sharper analysis of the tame exponents. Optimality below always means the least **divisibility** constant, or equivalently an lcm; it does not mean that one example must realize the entire constant.

The root for relative local source paths in this document is **C:/Users/14192/AI4Math/Rethlas/agents/genera**
The source inventory, with SHA-256 hashes, is source_manifest.md (source: `C:/Users/14192/AI4Math/Rethlas/agen`
The two supplied local inputs are the one-fiber problem and assumptions (source: `C:/Users/14192/AI4Math/Rethlas/`
and the verified family blueprint (source: `C:/Users/14192/AI4Math/Rethlas/agents/generation/results/target_`
Their precise assertions used here are restated in Proposition prop:supplied-inputs. Each external paper result is stated below with its own public source URL and identifiers.

Definition 1 (Constants)**.** For a prime q and $g \geq 1$, define

$$\begin{aligned} e_q(g) &= \max\big(\{0\} \cup \{a \geq 1 : q^{a-1}(q-1) \leq 2g\}\big), \\ \Lambda_g &= \prod_q q^{e_q(g)}. \end{aligned} \tag{1}$$

The product is finite, since $e_q(g) = 0$ for $q > 2g+1$. Thus $e_q(g)$ is the largest exponent a for which $\varphi(q^a) \leq 2g$. We retain

$$r(q,g) = \sum_{j \geq 0} \left\lfloor \frac{2g}{q^j(q-1)} \right\rfloor, \qquad M(2g) = \prod_q q^{r(q,g)}. \tag{1}$$

For a positive integer b, write $b_{p'} = b/p^{v_p(b)}$. Write $C_0(g,n)$ for the constant restricted to residue characteristic zero; $C_p(g,n)$, for prime $p > 0$, imposes residue characteristic p, allowing either characteristic of the valued field.

Proposition 2 (Supplied Inputs)**.** ***Statement.***

For the finite family in the question, set $F = K_v$, $I = I_v$, $N_i = I_{A_i,v}$, $H_i = I/N_i$, and $Q = I/\bigcap_i N_i$. Then

$$\begin{aligned} d_v(\mathcal{A}, T) &= |Q|, \\ Q &\hookrightarrow \prod_{i=1}^{n} H_i, \\ |H_i| &\mid M(2g). \end{aligned} \tag{2}$$

*The injection has surjective projections. In particular all these groups and degrees are finite. The minimum in (2) is an attained **total** degree of a finite separable extension of K, with one selected prolongation.*

For an abelian variety B/K, and a prime ℓ different from the residue characteristic,

$$I_{B,v} = \{\sigma \in I : \rho_{B,\ell}(\sigma) \text{ is unipotent on } V_\ell(B)\}. \tag{3}$$

This subgroup is open in I and normal in G_F. In residue characteristic $p > 0$, I has a normal pro-p subgroup P and procyclic prime-to-p quotient I/P. Every finite quotient H of I therefore has a normal Sylow p-subgroup W, the image of P, and cyclic prime-to-p quotient H/W. In residue characteristic zero, every finite quotient of I is cyclic.

Proof. These are precisely the established inputs used here, with their statements made explicit. Sources: paper_id = author-provided-one-fiber-draft; theorem_id = Section 1 of `data/target_rerun_20261004.md`, established one-fiber theorem; arXiv id = not supplied. For the family statements the source is `results/target_rerun_20261004/blueprint_verified.md`, specifically `lem:canonical-inertia-and-products`, `prop:exact-simultaneous-degree`, `prop:finite-set-bounds`, and `lem:ramification-structure`; paper_id = local-verified-family-blueprint; arXiv id = not applicable.

The aspects needed in the present extension can also be checked directly from those inputs. The diagonal map from I to $\prod_i H_i$ has kernel $\bigcap_i N_i$. Its projections are surjective. On the Tate module of the actual product $B = \prod_i A_i$, an inertial element is unipotent exactly when it is unipotent on each summand. Consequently $I_{B,v} = \bigcap_i N_i$. Applying the supplied one-fiber equality to this dimension-ng abelian variety proves the exact degree assertion, including total degree and the single chosen prolongation. The individual divisibility assertions use that theorem in dimension g.

Equation (3) and the ramification structure are used in their stated, previously verified forms. In particular the latter includes imperfect residue fields; it concerns inertia, not an identification of inertia index with ramification index. The topology in (3) is openness in I, not necessarily in G_F. No assertion about a minimum local degree is substituted for (2).

□

Lemma 3 (Cyclotomic Order Bound)**.** ***Statement.***

If the finite inertia quotient $H = I/I_{B,v}$ of a dimension-g abelian variety contains an element of order q^a, for a prime q and $a \geq 1$, then

$$\varphi(q^a) \leq 2g. \tag{4}$$

In particular, if a Sylow q-subgroup of H is cyclic, then $v_q(|H|) \leq e_q(g)$.

Proof. We use the following external result, including its local definitions. For an abelian variety X over a discretely valued field, let $H = I/I_X$. After passing to the maximal unramified extension of the completion, choose a finite extension w where X is semistable. Let t, a be the toric and abelian ranks of the connected special Néron fiber X_w^0, and t_v, a_v those before this extension. Thus $t + a = \dim X$. Over an algebraic closure of the residue field, let T_w, T_v be the special tori and set $T = T_w/\operatorname{im}(T_v)$. The torus T has dimension $t - t_v$. For ℓ different from the residue characteristic, put W equal to the orthogonal complement of $V_\ell(X_v)/V_\ell(T_v)$ in $V_\ell(X_w)/V_\ell(T_w)$, with respect to a pairing from a polarization of X. Let S be the intersection of W with the corresponding integral Tate-module quotient. Then the pairing on W is nondegenerate, alternating and H-invariant, W and S are independent of the polarization, and the natural actions induce an injection

$$\begin{aligned} H &\hookrightarrow \operatorname{Aut}(T) \times \operatorname{Aut}(W, e) \\ \operatorname{Aut}(T) \times \operatorname{Aut}(W, e) &= \mathrm{GL}_{t-t_v}(\mathbb{Z}) \times \mathrm{Sp}_{2(a-a_v)}(\mathbb{Q}_\ell). \end{aligned} \tag{5}$$

The first projection is independent of ℓ, and the characteristic polynomial of every element in the second projection has integer coefficients independent of ℓ.

Source: A. Silverberg and Yu. G. Zarhin, *Subgroups of inertia groups arising from abelian varieties*, paper_id = Silverberg-Zarhin-1998-inertia; theorem_id = Theorem 5.2(i); arXiv id = alg-geom/9706002. Paper. The downloaded paper and extracted text are in `downloads/target_rerun_20261004/silverb` Section 5, including the proof, has been read. Its proof passes to torus and abelian Tate pieces; a kernel would consist of finite-order unipotent operators and is therefore trivial. The rational-character assertion is essential here. The theorem applies to arbitrary discretely valued fields, not merely number fields or good-reduction fibers. Its group H is the finite component quotient of inertial algebraic monodromy, exactly the group in (3), rather than the possibly infinite full Tate-module image.

Taking the direct sum of the two representations in (5) gives a faithful characteristic-zero representation whose elementwise characteristic polynomials belong to $\mathbb{Z}[x]$, of dimension

$$D = (t - t_v) + 2(a - a_v) \le t + 2a \le 2g. \tag{2}$$

Let $h \in H$ have order q^a. Its representing matrix also has this order. A finite-order matrix in characteristic zero is diagonalizable over an algebraic closure. All its eigenvalues have q-power order dividing q^a; since their orders have lcm q^a, at least one has order exactly q^a. The integer characteristic polynomial must therefore be divisible over $\mathbb{Q}[x]$ by $\Phi_{q^a}(x)$. Its degree is at least $\varphi(q^a)$. This proves (4). A cyclic Sylow group contains an element of its full order, giving the final assertion.

□

Lemma 4 (Tame And Wild Upper Bounds)**.** ***Statement.***

In residue characteristic $p > 0$,

$$\begin{aligned} |Q|_{p'} &= \mathrm{lcm}_{1\le i\le n}(|H_i|_{p'}), \\ v_p(|Q|) &\le nr(p,g), \\ |Q| \mid & \; p^{nr(p,g)}(\Lambda_g)_{p'}. \end{aligned} \tag{6}$$

In residue characteristic zero,

$$|Q| = \mathrm{lcm}_{1\le i\le n} |H_i| \mid \Lambda_g. \tag{7}$$

Proof. First let $p > 0$. Write $W \triangleleft Q$ and $W_i \triangleleft H_i$ for the respective normal Sylow p-subgroups. The projections $Q \to H_i$ map W onto W_i, since all are images of wild inertia P. The induced map

$$Q/W \longrightarrow \prod_i H_i/W_i \tag{3}$$

is injective: the inverse image in Q of its kernel embeds, under (2), into $\prod_i W_i$, hence is a p-group and is contained in W. Both the source and each factor are cyclic of order prime to p, and the factor projections are surjective. The intersection of the projection kernels is trivial. For a cyclic group the index of an intersection of subgroups is the lcm of their indices. This gives the equality in (6).

For $q \neq p$, a Sylow q-subgroup of H_i injects into the cyclic group H_i/W_i, so it is cyclic. Lemma lem:cyclotomic-order-bound gives $v_q(|H_i|) \le e_q(g)$, and the equality just proved gives the same bound for Q. Its p-exponent is at most $\sum_i v_p(|H_i|) \le nr(p,g)$ by (2). This proves (6).

If the residue characteristic is zero, Q itself is cyclic. Apply the cyclic intersection argument to its projections onto the H_i, and apply (4) to their Sylow subgroups. This proves (7). No relation between different fibers' Tate modules is assumed.

□

Lemma 5 (Faithful Twisting)**.** ***Statement.***

Let k be algebraically closed, $K = k((t))$, and B/k an abelian variety of positive dimension. Suppose a finite group H acts faithfully on B by origin-preserving automorphisms. For a continuous homomorphism $\psi : G_K \to H$, let B_ψ/K be the twist of B_K defined by this action. Then

$$I = G_K, \qquad I_{B_\psi, v} = \ker \psi. \tag{8}$$

For finitely many such twists, their common inertia quotient is the image of the combined map to the product of the corresponding groups.

Proof. The residue field k is algebraically closed, so the unramified Galois quotient of the complete field K is trivial and $I = G_K$. For $\ell \neq \mathrm{Char}\, k$, all ℓ-power torsion of B is defined over k; consequently the original Tate representation is trivial.

The action of $\mathrm{End}(B)$ on $T_\ell(B)$ is faithful. For completeness, the union of the ℓ-power torsion is Zariski dense in B. Indeed its Zariski closure is a closed subgroup. If its identity component had dimension $d < \dim B = b$ and there were c components, each intersection of a component with $B[\ell^m]$ would have at most ℓ^{2dm} points, whereas $B[\ell^m]$ has ℓ^{2bm} points. This is impossible for large

m. An endomorphism killing all these torsion points therefore vanishes. Applied to the difference of an automorphism and the identity, this proves faithfulness on the Tate module.

Under the usual identification after a splitting extension, the Tate action of B_ψ is the specified action of $\psi(G_K)$, with compatible torsor conventions. Its kernel is thus $\ker\psi$. A finite-order operator over $\mathbb{Q}_\ell$ is unipotent exactly when it is the identity. Equation (3) proves (8). Intersecting these kernels gives the assertion for several twists. The twists exist by finite Galois descent: an invariant ample bundle is obtained by tensoring the translates of an ample bundle, and the group law descends with the scheme.

□

Lemma 6 (One Family For Prescribed Twists)**.** ***Statement.***

Let K be an infinite field, let B/K be an abelian variety, and let $H \subseteq \mathrm{Aut}_K(B)$ be a finite group. Given any n H-torsors $D_1, \dots, D_n$ over K, there exist a smooth geometrically integral finite-type K-scheme S, an abelian scheme $\mathcal{B}/S$ with relative dimension $\dim B$, and exactly n chosen distinct points $s_i \in S(K)$, such that $\mathcal{B}_{s_i}$ is the twist of B by D_i.

Proof. The proof has two separate ingredients: descent of the constant family, and a rational-point argument that realizes each prescribed torsor as a fiber. The cited twisting result is used only in the second ingredient, to show that the relevant twisted open subset has a K-point.

View H as the finite **constant** group scheme; it is étale even when the characteristic divides its order. Take the regular representation $V = K[H]$, and let

$$U = V \setminus \bigcup_{1 \neq h \in H} V^h. \tag{4}$$

Every V^h is a proper linear subspace. Thus U is nonempty, smooth and geometrically integral, and H acts freely on it. For $H = 1$, the union is empty. Finite group quotients give a finite étale H-torsor $U \to S_0 = U/H$. More explicitly $V \to \mathrm{Spec}\, K[V]^H$ is finite; removing the image of the invariant closed nonfree locus gives this quotient. Smoothness descends along its étale surjection, and geometric integrality follows from the geometrically integral cover. In particular S_0 is a scheme of finite type, not a moduli stack.

Descend $B \times_K U$ along this torsor using the given action on B; denote the descended abelian scheme by $\mathcal{B}_0/S_0$. Here and below a right torsor and a left action are combined with the standard contracted-product convention. For existence as a scheme, one can start with an ample line bundle L on B and take $\bigotimes_{h\in H} h^*L$; reindexing the factors supplies an H-linearization. This gives effective projective descent. The group law and origin descend because H acts by group automorphisms. Smoothness, properness, and geometrically connected abelian fibers descend as well.

Rational-point step. For each i, set

$$V_i = D_i \times^H V, \qquad U_i = D_i \times^H U.$$

We use the following result only at this point: if G is a linear algebraic group over K, D is a G-torsor over K, and W is a finite-dimensional G-representation, then the twist $D \times^G W$ is K-isomorphic to the affine space underlying W. Source: A. Duncan and Z. Reichstein, *Versality of algebraic group actions and rational points on twisted varieties*; paper_id = Duncan-Reichstein-versal-actions; theorem_id = Lemma 3.1; arXiv id = 1109.6093. Paper. In the present application this is the linear Galois-descent statement, equivalently Hilbert 90; the word "twist" denotes the contracted product. Applying this with $G = H$, $D = D_i$, and $W = V$ identifies V_i with an affine space over K. Since U is a nonempty H-stable open subset of V, U_i is a nonempty open subset of V_i. As K is infinite, $U_i(K) \neq \varnothing$. This is exactly where the quoted result is used: choose a point of $U_i(K)$.

Such a point is equivalent to an H-equivariant map of torsors $D_i \to U$ above some point $b_i \in S_0(K)$. The map identifies D_i with the pullback of $U \to S_0$ at b_i: after a separable extension it is an equivariant map between two H-torsors, hence an isomorphism. The fiber of $\mathcal{B}_0$ at b_i is therefore the required twist.

The preceding paragraph is the only place where the quoted twisted-affine-space result is used. Once the points b_i have been obtained, the rest is just a device for making the chosen points

distinct: take $S = S_0 \times_K \mathbb{A}^1_K$, pull back $\mathcal{B}_0$, and choose $s_i = (b_i, c_i)$ for distinct $c_i \in K$. These are exactly n chosen distinct rational points regardless of whether some torsors coincide. This construction needs no rationality assertion for U/H.

□

Lemma 7 (P Group Local Realization)**.** ***Statement.***

Let k be algebraically closed of characteristic $p > 0$ and $K = k((t))$. Every finite p-group R is the Galois group of a finite separable extension of K. Equivalently, $G_K = I$ admits a continuous surjection onto R. This applies in particular to $R = H^n$ for any finite p-group H.

Proof. We supply the central-extension argument so that independence is not inferred from existence of individual extensions. Artin–Schreier theory identifies

$$H^1(G_K, \mathbb{F}_p) = K/\{b^p - b : b \in K\}. \tag{5}$$

For any finite collection of distinct positive integers m prime to p, the classes t^{-m} are independent over $\mathbb{F}_p$. A nonzero linear combination has pole order equal to the largest such m. In contrast, if $v(b) < 0$, then $v(b^p - b) = pv(b)$, and if $v(b) \geq 0$, then $b^p - b$ has no pole. Thus $H^1(G_K, \mathbb{F}_p)$ is infinite-dimensional.

Also

$$H^2(G_K, \mathbb{F}_p) = 0. \tag{9}$$

Here is the complete external vanishing statement used: for an affine scheme X of characteristic p, $H^j_{\text{ét}}(X, \mathbb{Z}/p\mathbb{Z}) = 0$ for all $j \geq 2$. Source: paper_id = Stacks-Project; theorem_id = Lemma 59.63.1(1), Tag 0A3K; arXiv id = not applicable. Statement and proof, saved to `downloads/target_fixed_cardinality_20261006/stacks-0A3K.html`. The proof, read on that page, uses the Artin–Schreier exact sequence $0 \to \mathbb{F}_p \to \mathbb{G}_a \xrightarrow{F-1} \mathbb{G}_a \to 0$ and the vanishing of higher étale cohomology of the structure sheaf on an affine scheme. For $X = \operatorname{Spec} K$, étale cohomology is continuous Galois cohomology with the indicated trivial discrete coefficient module. This gives (9).

Induct on $|R|$. The trivial group is immediate. A nontrivial finite p-group has a central subgroup $C \simeq \mathbb{F}_p$. By induction choose an epimorphism $\bar{\psi} : G_K \twoheadrightarrow R/C$. The obstruction to lifting it to R is the pullback of the central-extension class in $H^2(R/C, C)$. It vanishes by (9), so a continuous lift $\psi : G_K \to R$ exists. Concretely, choose a section of $R \to R/C$. Its multiplication defect is a C-valued two-cocycle; vanishing of its pullback means it is the coboundary of a continuous one-cochain, and adjusting the section by this cochain gives the asserted homomorphism.

Let $J = \psi(G_K)$. If $J = R$, we are done. Otherwise $CJ = R$, while $C \cap J = 1$, because C has prime order. Since C is central, this identifies $R = C \times J$, with $J \simeq R/C$. Choose a character $\chi : G_K \to C$ that does not factor through $\bar{\psi}$; one exists because the space of characters factoring through the finite group R/C is finite-dimensional, whereas $H^1(G_K, C)$ is infinite-dimensional. Its restriction to $\ker \bar{\psi}$ is nonzero and hence surjective. Therefore

$$(\chi, \bar{\psi}) : G_K \longrightarrow C \times (R/C) \simeq R \tag{6}$$

is surjective. This finishes the induction. Its open kernel fixes a finite Galois, hence separable, extension with the required group. Finally $G_K = I$ by algebraic closedness of the residue field, so this is an inertia realization.

□

Lemma 8 (Maximal Wild Automorphisms)**.** ***Statement.***

For every prime p and $g \geq 1$, over an algebraically closed field k of characteristic p there exists a dimension-g abelian variety B with a faithful finite p-subgroup

$$H \subseteq \operatorname{Aut}_k(B), \qquad |H| = p^{r(p,g)}. \tag{10}$$

Proof. Suppose first that p is odd. The smooth projective model of

$$D: \quad y^2 = x^p - x \tag{7}$$

is a hyperelliptic curve of genus $h = (p-1)/2$: the polynomial on the right has derivative -1, has p distinct roots, and has odd degree p. The double cover of the x-line has these p branch points and the point at infinity. Riemann–Hurwitz gives the genus. Translation $x \mapsto x+1$, $y \mapsto y$, has order p and fixes the unique point at infinity. It acts faithfully on $J = \mathrm{Jac}(D)$. Indeed, for a pointed smooth curve of positive genus the Abel–Jacobi map $P \mapsto [P - O]$ is an embedding; an automorphism fixing O and acting trivially on its Jacobian must fix every point of the curve.

Put $m = \lfloor g/h \rfloor = \lfloor 2g/(p-1) \rfloor$, and take

$$\begin{aligned} B &= J^m \times E^{g-mh}, \\ H &= (C_p)^m \rtimes P_m. \end{aligned} \tag{8}$$

where E/k is any elliptic curve and P_m is a Sylow p-subgroup of the permutation group on m letters. The first factor acts independently on the J-factors; P_m permutes those factors; all act trivially on the padding. This action is faithful. For $m = 0$, use $B = E^g$ and the trivial group. Legendre's formula gives

$$\begin{aligned} v_p(|H|) &= m + v_p(m!), \\ &= \sum_{j \geq 0} \left\lfloor \frac{m}{p^j} \right\rfloor, \\ &= \sum_{j \geq 0} \left\lfloor \frac{2g}{p^j(p-1)} \right\rfloor. \end{aligned} \tag{11}$$

The last equality uses $\lfloor \lfloor x \rfloor / d \rfloor = \lfloor x/d \rfloor$ for an integer $d \geq 1$.

For $p = 2$, use the elliptic curve

$$E: \quad y^2 + y = x^3, \tag{9}$$

with its point at infinity. Let $\mathbb{F}_4 \subset k$. There are eight pairs

$$(a, b), \qquad a \in \mathbb{F}_4, \quad b^2 + b = a^3, \tag{10}$$

and the maps

$$u_{a,b}(x, y) = (x + a, y + a^2 x + b) \tag{12}$$

are automorphisms of E fixing infinity. Substitution verifies the equation, using $a^4 = a$. Their composition is

$$u_{a,b} u_{c,d} = u_{a+c,\, b+d+a^2 c}. \tag{11}$$

For $a = 0$, they are the identity and $z : (x, y) \mapsto (x, y+1)$. For $a \neq 0$, their square is z, since $a^3 = 1$. Thus the eight maps form a group H_0 of order eight (in fact Q_8); the description itself is sufficient, without using a classification of automorphism groups. These origin-preserving elliptic-curve automorphisms act faithfully.

On $B = E^g$, let $H = H_0^g \rtimes P_g$, where P_g is a Sylow 2-subgroup of the permutation group on g letters. Its action is faithful by the same independent-factor argument. Its order has exponent

$$v_2(|H|) = 3g + v_2(g!) = 2g + g + \sum_{j \geq 1} \left\lfloor \frac{g}{2^j} \right\rfloor = r(2, g). \tag{12}$$

This proves (10) for every prime.

□

Proposition 9 (Sharp Wild Exponent In One Family)**.** ***Statement.***

For every prime p, $g \geq 1$, and $n \geq 1$, there is a family satisfying the question over $K = \overline{\mathbb{F}}_p((t))$ with exactly n chosen distinct rational fibers and

$$\begin{aligned} Q &\simeq H^n, \\ d_v(\mathcal{A}, T) &= p^{nr(p,g)}. \end{aligned} \tag{13}$$

where H is the group from (10).

Proof. Choose $B/k, H$ from Lemma lem:maximal-wild-automorphisms, for $k = \overline{\mathbb{F}}_p$. Lemma lem:p-group-local-realization applied to the whole group H^n, rather than separately to its factors, supplies a surjection

$$\psi = (\psi_1, \ldots, \psi_n) : G_K = I \twoheadrightarrow H^n. \tag{13}$$

For $H \neq 1$, the coordinate maps define connected H-torsors, and their joint splitting extension has Galois group H^n. The trivial case gives trivial torsors. Use Lemma lem:one-family-for-prescribed-twists for these n torsors and B_K. It gives one smooth geometrically integral base and one abelian scheme containing the twists at n distinct K-rational points. Lemma lem:faithful-twisting identifies the fiber kernels as $\ker \psi_i$, so their common quotient is H^n. Formula (2) proves (13).

There is also an explicit degree witness: the extension fixed by $\ker \psi$ has total degree $|H|^n$, makes all the twists constant, and hence gives good reduction. Since K is complete it has a single selected prolongation; the common-inertia lower bound in (2) proves minimality even among non-Galois extensions. Thus (13) realizes the required total degree, not just an abstract group order or a ramification-index lower bound.

□

Lemma 10 (Sharp Cyclic Automorphisms)**.** ***Statement.***

Let k be algebraically closed, q a prime different from $\operatorname{Char} k$*, and $a \geq 1$. For odd q, there is an abelian variety over k of dimension $\varphi(q^a)/2$ with an origin-preserving automorphism of order q^a. For $q = 2, a \geq 2$, there is one of dimension $2^{a-2} = \varphi(2^a)/2$. For order two one can use an elliptic curve with its automorphism $[-1]$.*

Proof. First suppose q is odd, put $m = q^a$, and fix a primitive m-th root of unity $\zeta \in k$. If $\operatorname{Char} k \neq 2$, use the smooth projective model

$$D_a : \quad y^2 = x^m - 1. \tag{14}$$

The polynomial is squarefree because $m \neq 0$ in k, and the hyperelliptic genus formula gives $g(D_a) = (m-1)/2$. If $\operatorname{Char} k = 2$, instead use

$$D_a : \quad y^2 + y = x^m. \tag{15}$$

This Artin–Schreier cover is étale over the affine x-line. It has degree two: x^m cannot equal $b^2 + b$ for $b \in k(x)$, since a pole of $b^2 + b$ at infinity has even order, whereas m is odd. At any place above infinity, let e be the ramification index. Then $v(x) = -e$, the equation forces $v(y) < 0$, and $2v(y) = -em$. Thus e is even and therefore equals two. The degree bound consequently leaves exactly one place over infinity; its residue field is k. With the normalized valuation there, $v(x) = -2$ and $v(y) = -m$. The function $u = x^{(m-1)/2}/y$ is a uniformizer. For the nontrivial deck transformation $\tau(y) = y + 1$,

$$v(\tau(u) - u) = v\left(\frac{x^{(m-1)/2}}{y(y+1)}\right) = m + 1. \tag{16}$$

This is the different exponent of the quadratic extension. Riemann–Hurwitz therefore gives $2g(D_a) - 2 = -4 + (m+1)$. There is no ramification elsewhere, giving the asserted genus.

In both cases $\sigma : (x, y) \mapsto (\zeta x, y)$ acts on D_a, and the quotient by its subgroup $\langle \sigma^{q^{a-1}} \rangle$ of order q is D_{a-1}, via $f : (x, y) \mapsto (x^q, y)$. Here D_0 is the analogous curve with exponent 1 and is rational. Let $J_a = \mathrm{Jac}(D_a)$, and consider the connected image

$$B_a = \mathrm{im}\big([q] - f^* f_* : J_a \to J_a\big). \tag{14}$$

The image of an abelian variety homomorphism is an abelian subvariety. The pullback and norm identities on divisor classes are

$$\begin{aligned} f_* f^* &= [q], \\ f^* f_* &= \sum_{j=0}^{q-1} \sigma^{j q^{a-1}}. \end{aligned} \tag{15}$$

They hold for the quotient map, including at ramified points, with the ramification multiplicities in pullback. On rational Tate modules the operator $q^{-1} f^* f_*$ is consequently the averaging idempotent onto the subgroup invariants. Its image is $V_\ell(f^* J_{a-1})$, of dimension $2g(D_{a-1})$; the identity $f_* f^* = [q]$ shows that f^* has finite kernel. Thus its complementary idempotent has image $V_\ell(B_a)$, and

$$\dim B_a = g(D_a) - g(D_{a-1}) = \frac{q^a - q^{a-1}}{2} = \frac{\varphi(q^a)}{2}. \tag{17}$$

Choose $\ell \neq \mathrm{Char}\, k$. On $V_\ell(J_a)$ the finite-order operator σ is diagonalizable over $\overline{\mathbb{Q}}_\ell$. A q^a-th root eigenvalue survives the complementary idempotent precisely when its q^{a-1}-st power is not 1, equivalently when it is a primitive q^a-th root. Since B_a has positive dimension, the restriction of σ to B_a has order exactly q^a. This proves the odd-prime assertion in every allowed characteristic, including characteristic two.

For $q = 2, a \geq 2$, put $m = 2^{a-1}$, take a primitive 2^a-th root ζ, and use

$$D : \quad y^2 = x^{m+1} - x. \tag{18}$$

Here the characteristic is not two. The right-hand side has a simple root at zero, and at a root with $x^m = 1$ its derivative is $m \neq 0$. Thus the smooth projective model has genus $m/2 = 2^{a-2}$. The map

$$\alpha(x, y) = (\zeta^2 x, \zeta y) \tag{19}$$

preserves its equation, since $\zeta^{2m} = 1$, has order exactly 2^a, and fixes the unique point at infinity. The pointed Abel–Jacobi embedding used in Lemma lem:maximal-wild-automorphisms shows that its induced automorphism on the Jacobian has the same order. Finally $[-1]$ on any elliptic curve has order two in the permitted characteristics.

□

Proposition 11 (Sharp Tame Exponents)**.** ***Statement.***

Fix $g, n \geq 1$ and a prime q with $e_q(g) > 0$. In every prescribed residue characteristic different from q, including zero, some family in the question with exactly n chosen distinct rational points has degree

$$q^{e_q(g)}. \tag{16}$$

Proof. Write $a = e_q(g)$, and let k be algebraically closed of the prescribed characteristic: take $\overline{\mathbb{F}}_p$ for positive characteristic and $\mathbb{C}$ for zero. The previous lemma provides an abelian variety B_0/k of dimension at most g with an automorphism of order q^a. For $q = 2$, note that $g \geq 1$ ensures $a \geq 2$, so the dimension formula $\varphi(2^a)/2$ applies. Pad with a product of elliptic curves, on which the cyclic group acts trivially, to obtain a dimension-g abelian variety B/k with a faithful C_{q^a}-action.

Over $K = k((t))$, the extension obtained from $z^{q^a} = t$ is separable, Galois and totally ramified of degree q^a: the polynomial is Eisenstein and all its root ratios are in k. Twist B_K by its cyclic torsor. By (8) this abelian variety A/K has finite inertia quotient C_{q^a}, hence $d_v(A) = q^a$. Take the constant family $A \times_K \mathbb{A}^1_K$, with any n distinct K-points. Their inertia kernels coincide, giving

(16). Thus padding the number of fibers does not require extra dimension or a change in the valued field.

□

Proposition 12 (Exact Constants And Comparisons)**.** ***Statement.***

The exact constants are

$$\boxed{C_0(g,n) = \Lambda_g},$$
$$\boxed{C_p(g,n) = p^{nr(p,g)}(\Lambda_g)_{p'} \quad (p>0)}. \tag{17}$$

and

$$\boxed{C(g,n) = M(2g)^n = \operatorname{lcm}\big(C_0(g,n), \{C_p(g,n) : p \text{ prime}\}\big)}. \tag{18}$$

The earlier positive-residue-characteristic bound $p^{nr(p,g)}M(2g)_{p'}$ is sharp exactly when $p = 2, g = 1$; otherwise (17) is strictly smaller. In residue characteristic zero, $\Lambda_g < M(2g)$ for every $g \geq 1$, and the exact constant is independent of n.

Proof. The upper bounds in (17) are (6) and (7), together with the exact degree formula (2). Proposition prop:sharp-wild-exponent-in-one-family forces the full factor $p^{nr(p,g)}$ in any constant for residue characteristic p. Proposition prop:sharp-tame-exponents forces each $q^{e_q(g)}$ for $q \neq p$ in that same residue characteristic, and each prime factor of Λ_g in residue characteristic zero. These examples satisfy all family hypotheses, including exactly n distinct selected rational points. Since distinct prime-power requirements combine by lcm, (17) follows.

Every datum has residue characteristic zero or a prime, giving the lcm equality in (18). For each prime q, the residue-characteristic-q examples (13) require $q^{nr(q,g)}$ in this lcm. Conversely (2) gives $d_v(\mathcal{A}, T) \mid M(2g)^n$ in all characteristics. These matching exponent bounds prove (18). In particular the universal product upper bound really is optimal, although the tame factors in a fixed residue characteristic are much smaller. The constructions establish sharpness prime by prime, without claiming that a single example has degree $M(2g)^n$.

For each prime q, write $a = e_q(g)$. If $a > 0$, then for $0 \leq j < a$ the defining inequality $q^{a-1}(q-1) \leq 2g$ gives $\lfloor 2g/(q^j(q-1)) \rfloor \geq q^{a-1-j}$. Hence $r(q,g) \geq 1 + q + \cdots + q^{a-1} \geq a$; for $a = 0$ the inequality is immediate. In particular $\Lambda_g \mid M(2g)$.

For the strictness assertions, $r(2,g) = 3g + v_2(g!) \geq 3g$, whereas $e_2(g) = 1 + \lfloor \log_2(2g) \rfloor \leq 2g$. Thus $r(2,g) > e_2(g)$ for every $g \geq 1$. Together with the preceding divisibility this proves $\Lambda_g < M(2g)$, and proves strict improvement in the $p > 0$ bound whenever $p \neq 2$. When $p = 2$ and $g \geq 2$,

$$r(3,g) \geq g > 1 + \lfloor \log_3 g \rfloor = e_3(g), \tag{20}$$

where the strict inequality follows from $3^{g-1} > g$. Hence there is strict improvement in this case as well. For $g = 1, p = 2$, $M(2)_{2'} = (\Lambda_1)_{2'} = 3$, giving equality of the two bounds.

As a check, $\Lambda_1 = 12$ and $M(2) = 24$, and for every $n \geq 1$,

$$\begin{aligned} C_0(1,n) &= 12, \\ C_2(1,n) &= 3 \cdot 2^{3n}, \\ C_3(1,n) &= 4 \cdot 3^n, \\ C_p(1,n) &= 12 \quad (p \geq 5), \\ C(1,n) &= 24^n. \end{aligned} \tag{21}$$

At $n = 1$, equation (18) is exactly the supplied optimal one-fiber theorem. It does not turn its all-characteristic optimum into a fixed-characteristic claim.

□

Theorem 13 (Exact fixed-cardinality constants)**.** *Let $g, n \geq 1$ and consider the fixed-cardinality family setting above. The least divisibility constants in residue characteristic zero, in residue characteristic $p > 0$, and with no restriction on residue characteristic are respectively*

$$\begin{aligned} C_0(g,n) &= \Lambda_g, \\ C_p(g,n) &= p^{nr(p,g)}(\Lambda_g)_{p'}, \\ C(g,n) &= M(2g)^n. \end{aligned} \tag{22}$$

In residue characteristic zero the exact constant is independent of n and satisfies $\Lambda_g < M(2g)$. For fixed positive residue characteristic, the bound $p^{nr(p,g)}M(2g)_{p'}$ is sharp exactly when $p = 2$ and $g = 1$; otherwise the exact constant above is strictly smaller.

Proof. The preceding propositions determine the constants for every pair $g, n \geq 1$ in the original setting:

$$\begin{aligned} C(g,n) &= M(2g)^n, \\ C_p(g,n) &= p^{nr(p,g)} \prod_{q \neq p} q^{e_q(g)}, \\ C_0(g,n) &= \prod_{q} q^{e_q(g)}. \end{aligned} \tag{23}$$

where $e_q(g) = \max(\{0\} \cup \{a \geq 1 : \varphi(q^a) \leq 2g\})$.

Thus the universal all-characteristic constant $C(g,n) = M(2g)^n$ is optimal. For fixed $p > 0$, the wild exponent cannot be improved even when all n fibers lie in one abelian scheme: Proposition prop:sharp-wild-exponent-in-one-family constructs precisely such a family with degree $p^{nr(p,g)}$. The sharp replacement of the tame part is $(\Lambda_g)_{p'}$, by the cyclic Sylow constraint and the constructions in Proposition prop:sharp-tame-exponents. The former bound at fixed p is sharp only in the elliptic residue-characteristic-two case.

In residue characteristic zero gives $C_0(g,n) = \Lambda_g$ for every n, and this is strictly smaller than $M(2g)$ for every $g \geq 1$. For sharpness, the wild construction uses the finite quotient H^n of $G_{\overline{\mathbb{F}}_p((t))}$ and the smooth scheme $(U/H) \times \mathbb{A}^1$ to realize the n prescribed twists at distinct rational points. The tame construction uses a cyclic Kummer extension and a constant family with n distinct points. Each example has relative dimension exactly g, and its common inertia quotient proves the asserted minimum total degree at the same prolongation.

The all-characteristic constant follows by taking the lcm over residue characteristics. The exponent $nr(q,g)$ is forced at prime q by examples of residue characteristic q, while $M(2g)^n$ bounds every degree. At $n = 1$ this gives the supplied optimum $M(2g)$. No single example attaining all prime factors at once is required or asserted. These are exact constants, with no unresolved sharpness or common-family realization gap.

□